

Radioastronomy

Fortgeschrittenenpraktikum

HS1 Radioastronomy

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1. Introduction

The experiment HS1-Radioastronomy is there to give an overview for operating a radio telescope and to introduce the prerequisites such as characterising and calibrating the telescope before carrying out the experiments. The main experiment to be conducted is the observation of neutral hydrogen in the Milky Way to determine the rotational velocity in order to derive the mass distribution.

1.1. Radioastronomy

Radio astronomy is the study of electromagnetic radiation in the radio wavelength range, typically from a few MHz to several hundred GHz. Compared to optical astronomy, radio waves are much less affected by interstellar dust and gas, which allows astronomers to observe cold and dense regions of space that are often hidden in the optical regime. Radio telescopes usually consist of large parabolic antennas that collect incoming radiation and direct it to a receiver. The receiver amplifies the weak signal and converts it into an electrical voltage, from which the intensity, spectral distribution, and temperature of the observed source can be determined.

A central quantity in radio astronomy is the brightness temperature. In the Rayleigh-Jeans regime, which is valid for most radio astronomical observations, the spectral intensity is approximately

$$I_\nu \approx \frac{2k_B T_B \nu^2}{c^2},$$

where I_ν is the spectral intensity, k_B is the Boltzmann constant, T_B is the brightness temperature, ν is the frequency, and c is the speed of light. This relation shows that the observed radio signal is directly linked to the temperature of the emitting medium. This makes radio astronomy a powerful method for studying cold molecular clouds, ionized gas regions, and neutral hydrogen structures in the Milky Way and beyond.

One of the most important spectral lines in radio astronomy is the 21 cm line emitted by neutral hydrogen.

1.2. The 21 cm Hydrogen Line

The hydrogen atom consists of a proton and an electron. In the ground state, the proton and electron spins can be oriented either parallel or antiparallel to each other. This gives rise to a hyperfine splitting of the ground state: the antiparallel configuration has a slightly lower energy than the parallel configuration. A transition from the higher-energy state to the lower-energy state leads to the emission of a photon with a frequency of approximately

$$\nu \approx 1420.4 \text{ MHz},$$

which corresponds to a wavelength of

$$\lambda \approx 21.1 \text{ cm}.$$

This transition is therefore called the 21 cm line.

This line is of great importance in astronomy because hydrogen is the most abundant element in the interstellar medium. Neutral hydrogen emits strongly at 21 cm, allowing astronomers to map the distribution of gas in galaxies. The 21 cm line is especially useful because it is only weakly absorbed by interstellar dust, so it can reveal regions that are difficult or impossible to observe in the optical range.

The 21 cm line also enables the measurement of Doppler shifts. If the hydrogen gas is moving relative to the observer, the frequency of the emission is shifted. From this shift, the radial velocity of the gas can be determined. This makes it possible to study galactic rotation, gas dynamics, and the mass distribution of galaxies. For this reason, the 21 cm line is one of the most important tools in modern radio astronomy and is fundamental for studying neutral hydrogen in the universe.

2. Observation and Evaluation

2.1. The KRT3 telescope

2.2. Weather Conditions

The experiment was carried out on 07.09.2026, starting at around 12:35. The weather was stable throughout the day with partial cloud coverage and temperatures around 19°.

2.3. Calibration

As the receiver receives radiation not only from the source T_S but also from the environment T_h . In addition to those two external sources of radiation, the antenna also picks up on the radiation it's own system emits T_{sys} . This means the measured power of the receiver, represented by the antenna temperature, can be broken down the following way:

$$T_A = T_S + T_h + T_{sys} \quad (1)$$

It is difficult to separate the pure system temperature and the background radiation, so these temperatures are usually combined into an effective system temperature:

$$T'_{sys} = T_{sys} + T_h \quad (2)$$

The effective system temperature now includes all unwanted contributions. This means that to know the contribution of a source, one must know the effective system temperature. Determining the effective system temperature is calibrating the telescope by doing a 'On-Off' measurement through measuring a source with a known flux density and then measuring a piece of empty sky. The KRT3 however, can only measure the sun as a source with high enough flux density but the sun is not a constant source which is why a noise generator needs to be used.

The noise generator injects a signal directly into the antenna horn. The calibration temperature $T_{cal} = 170\text{ K}$ is the expected antenna temperature that is observed additionally when the noise generator is switched on.

When assuming the sun to be a point-like source ?? can be used in the form of:

$$T_b \Omega_s = T_A \Omega_A \quad (3)$$

with T_b being the brightness temperature, Ω_s being the solid angle of the sun, T_A being the antenna temperature and Ω_A being the solid angle of the antenna main lobe.

The goal is measuring the antenna temperature of the sun and converting it into a flux density. For this we need the following identities: the antenna theorem $A_e \Omega_A = \lambda^2$, the Rayleigh-Jeans law $B_\nu = 2kT_b/\lambda^2$, the flux density $S_\nu = B_\nu \Omega_s$ and the aperture efficiency $\eta = A_e/A_g = 0.5$ with A_g being the geometric antenna surface assumed to be a circle.

$$T_b \Omega_s = T_A \Omega_A \quad (4)$$

$$\Omega_A = \frac{\lambda^2}{A_e}, \Omega_s = \frac{S_\nu}{B_\nu}, B_\nu = \frac{2k_B T_b}{\lambda^2} \quad (5)$$

$$S_\nu = \frac{T_A \cdot 2k_B}{A_e} = f(T_A, A_e) \quad (6)$$

$$A_e = \pi r^2 \cdot \eta = 1,5^2 \pi \cdot 0,5 = \pi 1,125 \text{ m}^2 \quad (7)$$

$$\Rightarrow T_A = \frac{S_\nu A_e}{2k_B} = \frac{S_\nu \pi 1,125 \text{ m}^2}{2k_B} \quad (8)$$

2.3.1. Determination of the system temperature

2.3.2. Background Radiation

2.4. Determination of the angular resolution of the KRT3

The expected half-width is $q = 1.02\lambda/D = \dots$ with the wavelength $\lambda = \dots$ and the antenna diameter $D = 3\text{ m}$.

2.5. Determination of the Sensitivity of the instrument

To determine the theoretical sensitivity of the instrument, the weakest detectable flux density ΔS_ν for a given system temperature T'_{sys} can be given by the radiometer equation:

$$\Delta S_\nu = \frac{2k}{A_e} \frac{C_s T'_{sys}}{\sqrt{\Delta\nu\tau}} \quad (9)$$

where $C_s \approx 2$ is a dimensionless instrumental constant and τ the integration time. With Gleichung 9 the limiting sensitivity in a single channel with $\Delta\nu = 61$ kHz and an integration time of $\tau = 1$ s can be calculated to $\Delta S_\nu = \dots$

The measurement is carried out with the radio source Vir A with an elevation of $\dots^\circ$ at the time of measurement. The necessary integration time is calculated to be $\tau = \dots$, taking the flux density at $\nu = 1420,4$ MHz to be $S_\nu = 211$ Jy $\pm$ 11 Jy after Baars et al. (1977).

2.6. Observation of neutral hydrogen in the Milky Way

The preamplifier has a frequency range of 1398.00-1460.56 MHz divided into 1024 channels. The frequency resolution of a single channel (the bandwidth) is calculated ... As we observe the H1 line in a single channel, the velocity resolution is ... If we assume a radial velocity of 0 km/s

3. Results

3.0.1. Determination of the system temperature

3.0.2. Background Radiation

3.1. Determination of the angular resolution of the KRT3

3.2. Determination of the Sensitivity of the instrument

3.3. Observation of neutral hydrogen in the Milky Way

4. Discussion

5. Summary

Literatur

Abbildungsverzeichnis

Anhang A Tabellen

Tabelle 1: Messdaten des Dipol-Scans zur Bestimmung der kinetischen Energie (Schirm 3).

Dipolstrom I [A]	Vertikale Position y [mm]
2,371	-2,8116
2,278	-4,0231
2,181	-5,4668
2,076	-7,0907
1,963	-8,8247
1,868	-10,0697
1,768	-11,3757
1,678	-12,9589
1,568	-14,3459
1,477	-15,6398
1,372	-17,1793

Anhang B Lab book

```
1 # wetter
2
3 Datum: 07/09/26
4 um 12h35:
5 ![[Pasted image 20260907123451.png]]
6
7 teilweise bedeckter himmel
8 kein niederschlag
9
10 # 5.1.1 Calibration (system temp)
11 USB 1-5, 9-13
12
13 data structure in programm (RadioUniversePro)
14
15 | rep | dev | sb | on_tbi | off_tbi | cal_tbi | flux | SEFD |
16 | --- | ---- | --- | - | - | - | - | - |
17 | 1 | bbc01 | u | 12195,30 | 1001,40 | 2634,20 | 00,00 | 00,00 |
18  $T_{b} \cdot \Omega_{S} = T_{A} \cdot \Omega_{A}$ 
19  $\Omega_{A} = \frac{\lambda^2}{A_e}$ ,  $\Omega_{S} = \frac{S_{\nu}}{B_{\nu}}$ ,  $B_{\nu} = \frac{2k_B T_b}{\lambda^2}$ 
20  $S_{\nu} = \frac{T_A \cdot 2k_B}{A_e} \cdot f(T_A, A_e)$ 
21  $A_e = \pi r^2 \cdot \eta = 1,5^2 \pi \cdot 0,5 = 1,125 \pi \text{ m}^2$ 
22  $T_A = \frac{S_{\nu} A_e}{2k_B}$ 
23
24  $T_{\text{sys\_left}} = 98.76 \pm 22.54 \text{ K}$ 
25  $T_{\text{sys\_right}} = 107.80 \pm 31.65 \text{ K}$ 
26  $T_{\text{sun\_left}} = 1237.79 \pm 89.62 \text{ K}$ 
27  $T_{\text{sun\_right}} = 1302.59 \pm 119.16 \text{ K}$ 
28
29 With  $S_{\nu}$  of the Sun (aus SWC):
30 76 sfu(solar flux units):  $76 \cdot 10^{-11} \text{ Wm}^{-2} \text{ Hz}^{-1} = 76 \cdot 10000 \text{ Jy} = 760000 \text{ Jy}$ 
31  $\Rightarrow T_A = 972,753 \text{ K}$ 
32 - störung wegen rfi
33 # 5.2 angular resolution
34 USB 1-5, 9-13
35 position of the sun calculated by the control software
36
37 | Source | RA | DEC | AZ | EL | HA | Type | Note |
38 | ----- | ----- | ----- | ----- | ----- | ----- | ---- | ---- |
39 | Sun | 110410.1 | +055810.9 | +1684733.7 | +42 00 48.8 | -003253.9 | Star | East |
40 in ordner cscan ist der screenshot
```

```

41 in ordner FITS sind die daten (lon vs lat)
42
43 # 5.3 sensitivity of the instrument
44
45 antenna temp of VirA
46 BBC 3,11
47 shows least RFI influence
48
49
50 time needed for mapping:
51 28:35.4s (software)
52 29:27.8s (own timer)
53
54 screenshot auf ubuntu und nicht windows!
55 files bei ordner maps und in FITS als imageproJ01 VirA 01
56
57 TODO:
58 ## Theoretical Sensitivity
59 
$$\Delta S_{\nu} = \frac{2k}{A_e} \frac{C_S T'_{sys}}{\sqrt{\Delta \nu \tau}}$$

60 spectral flux density
61  $C_S \approx 2$ 
62  $T'_{sys, left} = 98.76 \pm 22.54 \text{ K}$ 
63  $T'_{sys, right} = 107.80 \pm 31.65 \text{ K}$ 
64  $A_e = 1.125 \pi$ 
65  $\tau = 1 \text{ s}$ 
66  $\Delta \nu = 61 \text{ kHz}$ 
67  $\Delta S_{\nu, left} = 6,248 \cdot 10^{-24} \frac{\text{W}}{\text{m}^2 \text{ Hz}} = 624,92 \text{ Jy}$ 
68  $\Delta S_{\nu, right} = 6,82 \cdot 10^{-24} \frac{\text{W}}{\text{m}^2 \text{ Hz}} = 682,02 \text{ Jy}$ 
69
70 Necessary integration time  $\tau$  to detected the selected source with the KRT 3 with a
    signal-to-noise ratio  $S/N = 10$  in each channel:
71 - Virgo A: Flux density  $= 211 \pm 11 \text{ Jy}$ , Radiogalaxy
72 - for 5 channel each side ( $\Delta \nu = 5 \cdot 61 \text{ kHz} = 305 \text{ kHz}$ )
73 -  $\frac{S}{N} = 10$ 
74 -  $\tau = \left( \frac{2k \cdot C_S \cdot T'_{sys}}{\Delta \nu \cdot A_e} \right)^2 \cdot \frac{S}{N}$ 
75 -  $\tau_l = 4,12 \text{ s}$ 
76 -  $\tau_r = 5,22 \text{ s}$ 
77 ## Determination of the uncertainty of radiant flux measurements
78
79 ### preliminary work
80  $S_{\nu} = \frac{P}{4 \pi d^2 \sqrt{2 \pi}} \sigma = 1263,16 \text{ Jy}$ 
81 für mobile radio on the moon (transmitting with 2W, 900MHz, FWHM:80kHz,  $\sigma = 34 \text{ kHz}$ ,
    recieved on earth
82
83  $S_{\nu} \propto \nu^{-\alpha}$ ,  $\alpha = -0,7$ 
84
85 bei  $\nu = 1420.4 \text{ MHz}$ :  $S_{\nu, CasA} = 1420 \pm 70 \text{ Jy}$ 
86  $\frac{1420,4^{-0,7} \text{ MHz}}{1420 \text{ Jy}} = c$ 
87  $S_{\nu, CasA} = \frac{900^{-0,7}}{c} \text{ Jy} = 1954,37 \text{ Jy}$ 
88
89  $T_A = \frac{S_{\nu} A_e}{2k B} = 1,82 \text{ K}$ , für  $\nu = 1420.4 \text{ MHz}$  again
90
91 TODO: Fehlerfortpflanzung
92 ### evaluation
93 bbc 3,11
94 ![[Pasted image 20260909124324.png]]
95 über ds9:
96 ![[Pasted image 20260909134422.png]]
97
98 ![[Pasted image 20260909134436.png]]
99
100  $\sigma_{mean} = 1455,83$ 
101  $\sigma_{std} = 212,206$ 
102  $\sigma_{scaled} = \sigma [K] = \frac{\sigma_{std}}{\sigma_{mean}} \cdot T'_{sys} = 15,71 \text{ K}$ 
103  $\sigma_{mean\_scaled} [K] = T_A (OFF) = 156938,474 \text{ K}$ 
104  $T_A (ON) = 3067 \cdot 107,8 \text{ K} = 330622,6 \text{ K}$ 

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105 ![[Pasted image 20260909134522.png]]
106 Antenna Temp of source  $T_{\text{s}} = \frac{T_{\text{A}}(\text{ON}) - T_{\text{A}}(\text{OFF}) + T'_{\text{s}}}{T'_{\text{sys}}} = 1612,17 \text{ K}$ 
107 Rauschen sehr viel größer als erwartet
108 vergleich mit theoretischer empfindlichkeit (600Jy, ca. 1K)
109 bei uns jtz 128 Kanäle, 10s integrierzeit
110 also eig sollte es noch besser sein( <1K)
111 aber bei uns viel einfluss von RFI -> ~15K std/rauschen
112 empfänger/verstärker sensibilität ist ein problem
113 termisches rauschen begrenzt nicht
114 aber besser als 1K kanns eig nicht werden
115
116  $\frac{S}{N}$  gesucht
117 - 25 karten für S/N=10
118 # 5.1.2 background radiation
119 ## moon
120 for moon: elevation 30 52' 32.0''
121 (in deg, min, sec)
122
123 for BCC 1-5,9-13
124 (as before for the sun)
125 even if quality is shit
126
127 span = 2(ELV -10 ) = 2(30 -10 )=2(20 )=40
128 but because azimuth at 268 03' 32.7'' chose slightly smaller span
129 -> span = 38
130
131 around 14.56
132
133 data in FITS: LON-MOON und LAT-MOON
134 +screenshot in cscan: CSCAN-MOON
135 ## virgo A
136 for Virgo A: at 15.28
137 elevation 47 56' 18''
138 azimuth= 196
139 span = 74
140
141 # Observation of neutral hydrogen in the Milky Way
142 ## frequency setting of the receiver
143
144 ### preliminary work
145
146 - What is the frequency resolution , i.e. the bandwidth of a single channel? Then
147   what is the velocity resolution when observing the H I line?
148 - In which channel would you expect the H I line at  $\nu_0 = 1420.4 \text{ MHz}$ , assuming a radial
149   velocity of  $0 \text{ km s}^{-1}$  ?
150 - You want to measure the radio continuum of a source. What frequencies are allowed
151   within the bandwidth so that the measurement is not be contaminated by the 21cm line.
152   Assume a maximum radial velocity of the H I of  $\pm 200 \text{ km s}^{-1}$  ;
153 - So which frequency bands should be selected in BBC Tools if you want to observe the H I
154   line once and the continuum once?
155 ## measurement
156  $R = R_{\text{0}} \cdot \sin(1)^\circ$ ,  $R_{\text{0}} = 8,5 \text{ kpc}$ 
157
158 |  $1^\circ$  [deg] |  $R$  [kpc] |  $v_{\text{max}}$  [ $\frac{\text{km}}{\text{s}}$ ] |  $v_{\text{R}}$  [ $\frac{\text{km}}{\text{s}}$ ] |
159 | ----- | ----- | ----- | -----
160 | 10 | 1,48 | 4,09 | - | -
161 | 20 | 2,91 | 4,0 | 70,70 | 14,50 | ...
162  $v_{\text{max}}$  grafisch bestimmt (schnittwert mit der orangenen  $\sigma$ -linie)

```

```

163 $v_{R}=v_{\max}+\omega_{0}R_{0}\cdot \sin(l)$
164 $\omega_{0}R_{0}=220 \frac{\text{km}}{\text{s}}$
165 $v_{\max,\text{err}}$ grafisch über blauen bereich bestimmt (schnittpunkt min-schnittpunkt max/2)
166 $\Delta l=\pm 2$ $
167 Fehlerfortpflanzung für R:
168 $\Delta R=\frac{\Delta R}{\Delta l}\cdot \Delta l=R_{0}\cdot \cos(l)\cdot \Delta l$
169 $\sigma_{R}=\sqrt{|R_{0}\cos(l)|\Delta l}$
170 $M(R)=0,234\cdot 10^{10}\left(\frac{v(R)}{\frac{100\text{km}}{\text{s}}}\right)^2\left(\frac{R}{\text{kpc}}\right)\cdot M_{\text{sun}}$
171 $M_{\text{sun}}=1,98\cdot 10^{30}\text{kg}$
172 ges: eingeschlossene Masse
173 ## distrib dark matter
174
175 $M_{\text{DM}}(R_0)=66411575447.22 \pm 3593349279.00$

```